

Course:	Object Oriented Programming	Semester:	II	Roll No:	
Division:		Batch:		Venue	
Date and time:	13/02/2026 (2:15 to 3 PM)	Name of Supervisor		Supervisor Sign:	

Debug the following program:

Instructions The given program contains **compiler errors, logical errors, syntax issues**. Your task is to: **Identify and explain the errors** in the program. **Provide the corrected code** in the specified format.

Scoring Criteria: Error Identification & Explanation: 1 mark per correct identification.
Correction: 1 mark per correct fix.

```

1 interface StudentInfo
2 {
3     static void displayInfo();
4 }
5 class Person extends StudentInfo
6 {
7     String name;
8     int rollNo;
9     static int totalStudents = 0;
10
11     Person(String name, int rollNo)
12     {
13         name = name;
14         rollNo = rollNo;
15         totalStudents++;
16     }
17
18     static void showDetails()
19     {
20         System.out.println("Name: " + name);
21         System.out.println("Roll No: " + rollNo);
22     }
23
24     public void displayInfo()
25     {
26         System.out.println("Student Information");
27     }
28 }
  
```

```

29
30 public class DebugDemo
31 {
32     public static void main(String args[])
33     {
34         Person p1 = new Person("Rahul", 101);
35         int marks[] = new int(5);
36         p1.showDetails();
37         StudentInfo.displayInfo();
38     }
39 }
  
```

Write your answers in the following table: (Any 5 out of six)

Line No	Error	Explanation	Corrected Code
3	Interface method declared as static without body	Interface methods cannot be static abstract without body	void displayInfo();
5	Wrong syntax of interface	implements keyword used for interface with class	class Person implements StudentInfo
13,14	Instance variable shadowing	Local variables hide instance variables	this.name = name; this.rollNo = rollNo;
18,20,21	Static method accessing non-static variable	Static methods cannot access instance variables directly; Remove static keyword from method	void showDetails ()
35	Wrong array declaration	Arrays use square brackets	int marks[] = new int[5];
37	Interface method cannot be called using interface name	Need object reference	p1.displayInfo();

Instructions The given program contains missing statements. Your task is to fill in the blanks with appropriate Java code.

Complete the following program:

```
1 class Circle {
2     double x,y;
3     double r;
4     Circle(double a, double b, double c){
5         x = a;
6         y = b;
7         r = c; } // 1 Marks
8     Circle(double c){
9         x = 0;
10        y = 0;
11        r = c; } // 1 Marks
12    Circle(Circle c){
13        x = c.x;
14        y = c.y;
15        r = c.z; } // 2 Marks
16    Circle(){
17        x = 0.0;
18        y = 0.0;
19        r = 1.0; } // 1 Marks
20    double circumference(){
21        return 2*3.14159*r; } // 1 Marks
22    double area(){
23        return (22/7)*r*r; } // 1 Marks
24 }
25
26 class Demonstration_39{
27     public static void main(String args[]){
28         Circle c1 = new Circle(3.0,4.0,5.0); // 1 Marks
29         Circle c2 = new Circle(5.0);
30         Circle c3 = new Circle(c1 or c2); // 1 Marks (c3 should
31             have data members exactly same as any existsting
32             object)
33         Circle c4 = new Circle();
34         System.out.println("Circumference Circle 1 " + c1.circumference
35             ());
36         System.out.println("Area Circle 1 " + c1.area()); // 1 Marks
37     } }
```